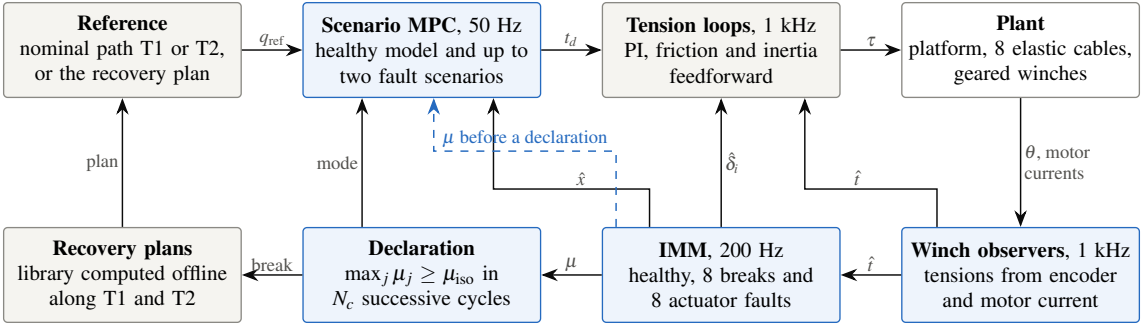
Reference
nominal path T1 or T2,
or the recovery plan
q_{ref}
Scenario MPC, 50 Hz
healthy model and up to
two fault scenarios
t_d
Tension loops, 1 kHz
PI, friction and inertia
feedforward
τ
Plant
platform, 8 elastic cables,
geared winches
θ, motor
currents
plan
mode
μ before a declaration
$\hat{x}$
$\hat{\delta}_i$
$\hat{t}$
Recovery plans
library computed offline
along T1 and T2
break
Declaration
$\max_j \mu_j \geq \mu_{\text{iso}}$ in
N_c successive cycles
μ
IMM, 200 Hz
healthy, 8 breaks and
8 actuator faults
$\hat{t}$
Winch observers, 1 kHz
tensions from encoder
and motor current